

Lumielle Academic Assistant

A local workflow from research topic to a reviewable Word manuscript

Research facts · Literature evidence · Chapter writing · Reviewable export

Lumielle Academic Assistant brings research-material management, literature analysis, outlining, chapter drafting, and Word export into one local desktop workflow. It assists with organization and drafting; it does not replace researcher judgment, source verification, or academic integrity requirements.

v1.1.0 — Unreleased · 2026

Writing assistance · Review all output

1. Purpose and limits

The application runs locally and stores project files on the machine. When an LLM service is called, selected content is sent to the provider configured by the user. Model output can omit, misread, or invent details and must be reviewed by the author.

- Useful for organizing a research framework, managing literature, drafting chapters, and preparing a Word manuscript.
- Not intended for unattended submission, specialist database searching, or treating generated text as verified fact.
- Keep backups of source material and drafts. Check important data, citations, methods, and conclusions against their sources.

2. Installation and first launch

Use the distribution folder that matches your operating system and language. Keep its launcher and bundled runtime files in place.

- On macOS, double-click Launch.command for English or 「一键启动.command」 for Chinese. On Windows, use Launch.bat or 「一键启动.bat」 respectively.
- First launch may prepare a Python environment and install dependencies. After the browser opens, enter the endpoint and API key in LLM Configuration and choose a model.
- The distribution's data/ directory stores local configuration, research notes, literature records, drafts, images, and templates. Back it up before changing installation files; ordinary errors are not a reason to delete it.
- Built-in Word presets are stored in format_presets/. The application version appears in the sidebar.

3. Recommended workflow

Set the topic and scope first, organize materials and structure, then draft and export chapter by chapter.

- Enter the global research topic and target total length, then configure the LLM service.
- Add source materials to the Research Foundation by uploading PDF, DOCX, or TXT files, or by pasting text. Review the extracted facts and filing.
- Import literature and inspect its profile and evidence. Bind applicable references to outline chapters.
- Generate and edit the logic outline; check chapter order, argument goals, length, and chart requirements.
- Draft individually or in a batch. Review facts, citations, and continuity before exporting with a FormatSpec and optional DOCX template.

4. Research Foundation and fact sections

The Research Foundation organizes source material as editable facts. Eight preset sections cover background, requirements, objectives, theory, methods, data, results, and constraints; custom sections are also available.

- Upload a source or paste text to extract facts in ordered chunks. Failed parsing does not import partial results. Scanned PDFs without a text layer need conversion to searchable text first.
- Edit each fact's title, content, tags, source, and chapter binding. Smart Research Inbox classifications are suggestions; review or change the section, tags, role, and writing-grounding setting before accepting.
- When writing grounding is enabled, the app selects relevant eligible facts rather than sending every fact to every chapter. Explicit bindings and global constraints receive priority. Disabled sections can still inform planning and outlining.
- Logic planning builds a bounded Research Planning Digest from saved facts in every section, including sections disabled for chapter grounding. When the budget is exceeded, balanced excerpts represent source material and mark omitted detail; late facts are not dropped merely because of their position.
- Context space is limited, so injected facts are selected and bounded. Unrelated or ambiguous material can add noise; curate and verify it before drafting.

5. Full-document literature analysis and evidence

Literature import supports text-bearing PDF, DOCX, TXT, and pasted text. Original files are stored in the literature-source directory; profiles and evidence are indexed separately.

- Extractable full text is processed in ordered chunks and synthesized into fields such as research question, theory, methods, sample, data, findings, limitations, and summary. Processing the full source does not mean the entire document is sent on every later generation request.
- Chunk extraction remains neutral to the project; final profile synthesis receives the current global research topic and Research Planning Digest. The rating measures relevance and usefulness to this project, while rigor, evidence strength, limitations, and source-supported timeliness are assessed separately.
- Whole-library quality and gap review uses structured profiles. Larger libraries are reviewed in ordered profile batches and then synthesized hierarchically rather than reduced to a fixed prefix.
- Evidence items record a claim, an exact source quotation, and a chunk locator. Only quotations found in the corresponding source chunk are retained; authors should still verify context and interpretation in the original source.
- Chapter writing ranks relevant evidence with Chinese or English terms and retrieves it only from literature bound to that chapter. A binding does not prove that the citation or interpretation is correct.
- Legacy literature records remain displayable. If an original source exists, it can be analyzed again; failure preserves the prior profile and evidence. An index cannot reconstruct a missing source document.
- Clearing literature records and deleting imported originals are separate choices. Source deletion is attempted only after explicit confirmation and is limited to managed files under `raw_files/`; partial failures are reported.

6. Outline, chapter memory, and Context Sandwich

Chapter generation combines the global topic and outline, upstream context, the current chapter, downstream written material or unwritten boundaries, enabled research facts, bound literature evidence, and writing requirements.

- The complete text of a written chapter is processed in ordered chunks to build a compact memory. Later generation uses this summary; it does not resend every previous chapter in full each time.
- The current chapter remains the focus. Downstream written content helps with continuity checks; unwritten sections provide boundary cues and should not be drafted ahead of schedule.
- Summarization and generation can still omit or misread information. Check key numbers, definitions, methods, results, and limitations. Context Sandwich is intended to reduce repetition and inconsistency risks; it cannot guarantee perfect recall or consistency.
- After a successful chapter rewrite, its memory is rebuilt from the new text. A failed rebuild does not replace the previous valid memory with a partial result.

7. Chapter writing and batch workbench

The writing module supports context review, single-chapter drafting and editing, length correction, and queued multi-chapter work.

- Before drafting, review the chapter title, description, target length, research facts, reference bindings, and context boundaries. The latest saved writing-style prompt is used on the next request.
- Batch tasks process chapters and show progress. Check configuration, network, and model responses before retrying a failed chapter; errors should not be treated as prose.
- Drafts, chapter summaries, reference bindings, charts, and images are separate data. Confirm the selected chapter before regenerating or deleting material.
- Charts are rendered locally only after their data specification is validated. Do not treat illustrative values as research results. Uploaded images can include captions.

8. FormatSpec and Word export

Word export uses data-only FormatSpec schema v1. Built-in and custom presets are validated against supported fields and ranges; presets do not carry Python code or file paths.

- General Academic uses A4 academic margins, page numbers and captions, and three-line tables. Legacy Compatible approximates the older margins and grid tables without adding page numbers or numbered captions by default; the presets produce distinct layouts.
- FormatSpec v1 supports page-number start, keep-with-next and page-break-before controls for each heading level, grid or three-line tables, table-caption enablement, and figure width as a percentage of printable width.
- Import, export, rename, and delete custom JSON presets. A model-generated preset is validated and shown as a preview; it is saved only after user confirmation.
- The natural-language format prompt is input for generating a FormatSpec. It affects Word export only after a generated specification is validated and saved or selected.

- Choose a FormatSpec, a DOCX template, or a template with FormatSpec. Template-only mode aims to retain existing page and paragraph styles; combined mode applies supported FormatSpec fields. Review the output for template-specific elements and pagination.
- Export orders references by first appearance and remaps chapter-local citation numbers to the document-wide registry. Verify in-text citations against the final reference list.
- Complex template fields, macros, special objects, and layout requirements outside the schema are not controlled by FormatSpec. Word-compatible applications may display fields and pagination differently.

9. Detection, rewriting, and logic review

AIGC detection, rewriting, and full-paper logic review are assistive tools; their results require author judgment.

- A detection score is not an academic-integrity finding and does not prove that text was written by AI.
- After rewriting, compare facts, numbers, terminology, citations, and tone. Do not change research meaning to chase a lower score.
- Verify logic-review findings against the manuscript and sources. The report does not replace advice from a supervisor, peer, or editor.
- API errors, empty responses, and invalid outputs should be reported as failures rather than saved as prose.

10. FAQ, data, and license

The app manages project files locally. When calling an LLM, prompts, selected source text, and API credentials are sent over the network to the configured provider.

- The API key is stored in local configuration and is also sent to the configured provider with API requests. Review that provider's privacy, retention, and training policies; do not submit materials you are not authorized to share.
- For an import error, check the supported format and whether a PDF has a text layer; scanned images are not OCR-processed. For model errors, check endpoint, key, model name, and network access.
- Ordinary project state is in the distribution's data/ directory. Back it up before modifying an installation or restoring a snapshot; do not share API keys or paper data unnecessarily.
- The project is open-source under the MIT License. Users are responsible for reviewing and appropriately using generated, detected, and formatted output.

Project identity



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Appendix A: Version History

Current development version: v1.1.0 (Unreleased). This version is not a formal release yet.

v1.1.0 — Unreleased

Added

- Added an eight-section Research Fact Layer with ordered chunk extraction, editable Smart Inbox suggestions, source information, tags, and explicit chapter-grounding controls.
- Added a bounded Research Planning Digest across all fact sections, including sections disabled for chapter grounding.
- Added full-document literature analysis for PDF, DOCX, TXT, and pasted text, with structured profiles and source-linked evidence; chapter retrieval is limited to bound literature.
- Added topic- and planning-aware literature profile synthesis; relevance ratings are separate from source-quality assessments.
- Added FormatSpec v1 support for grid/three-line tables, page-number starts, heading pagination controls, figure width, and distinct General Academic and Legacy Compatible presets.
- Updated the Chinese and English PDF user guides and unified the application version to 1.1.0.

Changed

- Chapter prompts can combine enabled research facts, bound literature evidence, the global outline, and surrounding chapter context.
- Planning digests cover facts across all sections; literature profiles use the current topic and planning context, and chapter evidence ranking supports Chinese and English terms.
- Large literature libraries are reviewed in ordered profile batches and hierarchically consolidated instead of being reduced to a fixed prefix.
- FormatSpec controls only strictly validated data fields; template behavior depends on the selected mode, and a natural-language format prompt must first produce a selected valid specification.
- README and manuals now describe complete-chapter memory extraction and compact-summary use without promising that models never forget or conflict.

Fixed

- Chapter memory no longer analyzes only the first 3,000 characters: long chapters are processed in order before synthesis, and failures preserve the old memory.
- Clearing literature records is separated from deleting imported originals; deletion is constrained to managed raw_files/ paths and reports partial failures.
- Failed literature analysis does not replace the prior valid profile or evidence; legacy records can be re-analyzed when their source file exists.
- Accepted Smart Inbox facts now persist the user-editable role field.
- Restored the manual's document-flow layout and Lumielle poster asset after a visual regression.

v1.0.1 — 2026-09-27 · Added / Changed / Fixed

- Added the canonical SemVer source version.py and displayed the application version in both locales.
- Added regression tests and Python 3.12 GitHub Actions CI.

- Drafts retain reference snapshots and mappings for citation export.
- Replaced model-generated chart code execution with validated chart specifications and trusted local rendering.
- Added ordered chunk parsing for complete Research Foundation documents and manuscript reverse engineering.
- Chapter prompts receive the global outline, upstream context, and downstream written content or boundaries.
- Saved custom style prompts are interpolated; English word counts use word tokens; English defaults initialize in prompts.json without replacing existing settings.
- Word export uses a global ordered citation registry and saved draft snapshots, with a legacy binding fallback.
- LLM failures, empty results, failed literature ratings, failed detection, and failed review blocks are kept out of normal saved content; prior valid outlines and reports are preserved on failure.
- Fixed macOS first-run environment creation using the bundled uv runtime and a pip-enabled environment.
- Removed non-commercial wording that conflicted with the MIT License; no core user workflow was removed.

v1.0.0 — 2026-08-26

Initial public release with Chinese and English desktop versions for macOS and Windows. Historical release dates are retained; v1.1.0 remains Unreleased until formally approved.